

02464

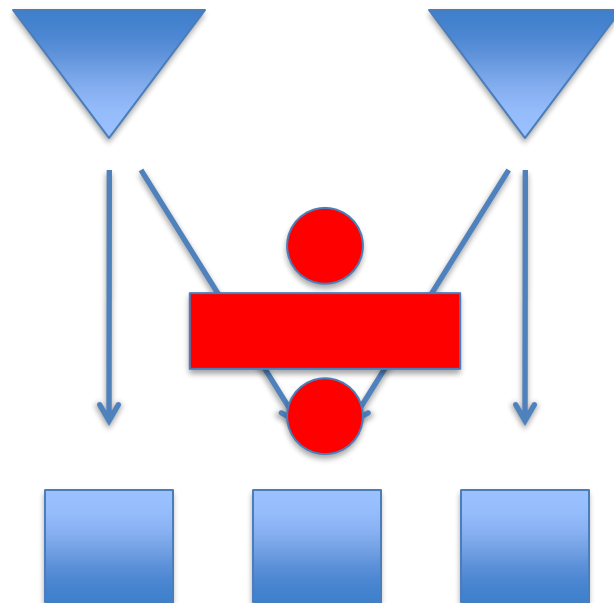
Kunstig Intelligens og Menneskelig Kognition

Linear encoding and PCA

A modest proposal

- The perceptual system is optimized to represent information relevant for survival
 - The dynamic range of neurons match the dynamic range of stimuli
 - Predators have binocular vision
 - Information is represented in an efficient way in the human brain

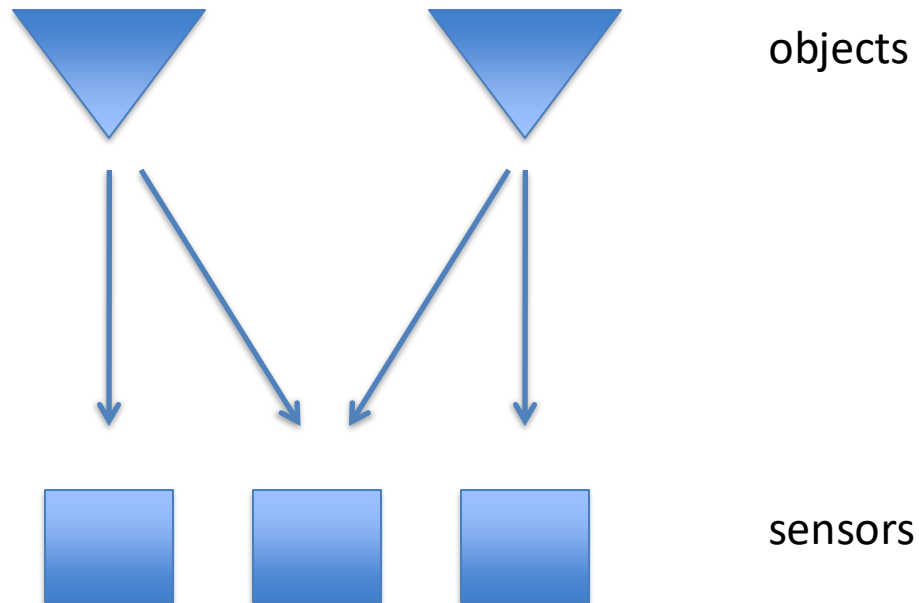
Efficient representations



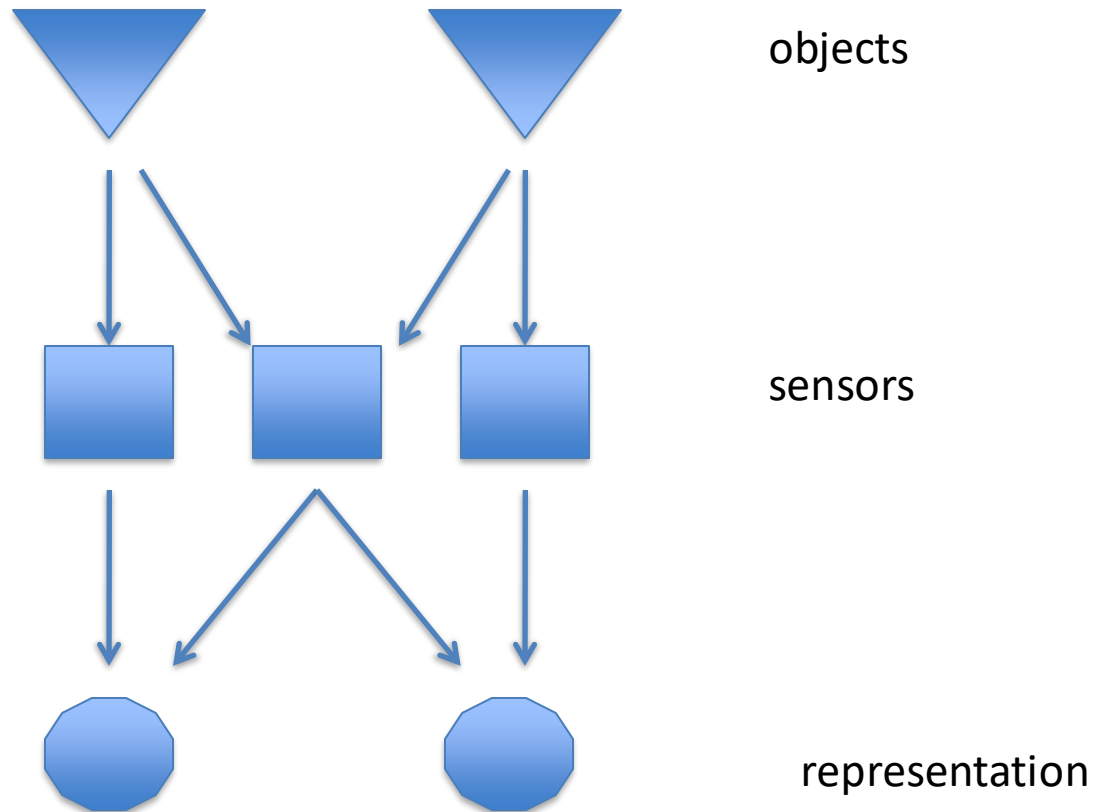
sensors

Efficient representation?

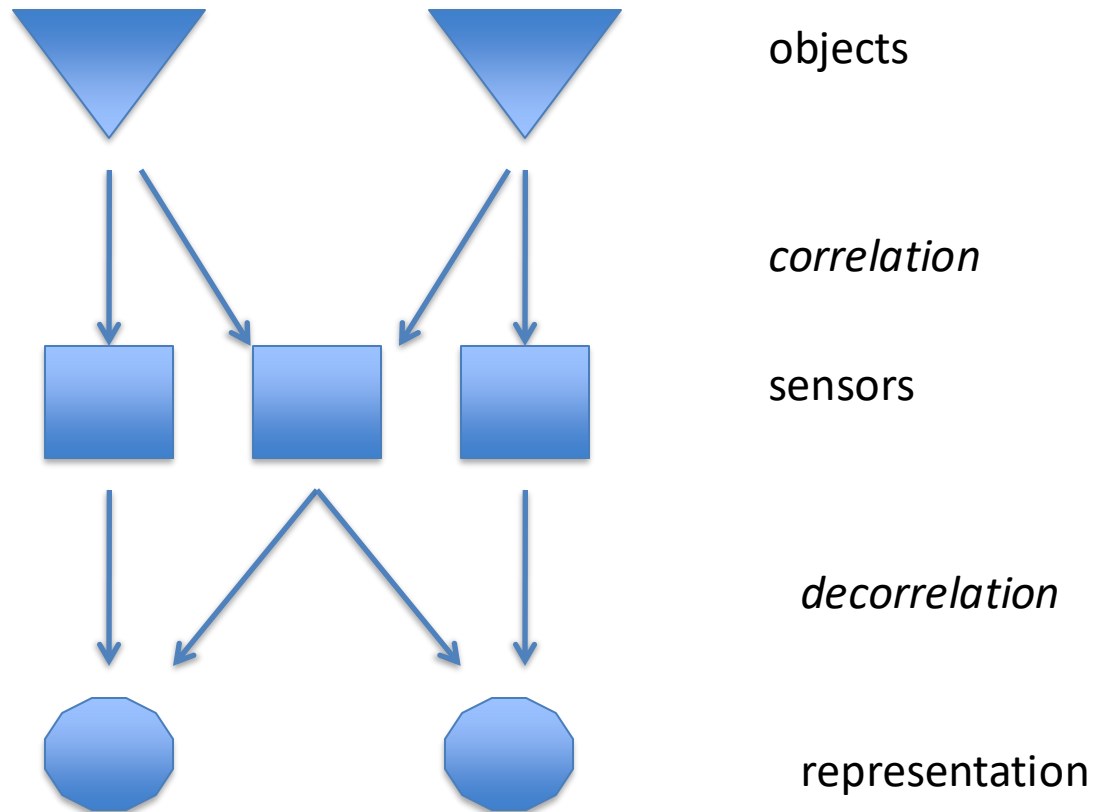
Efficient representations



Efficient representations



Efficient representations

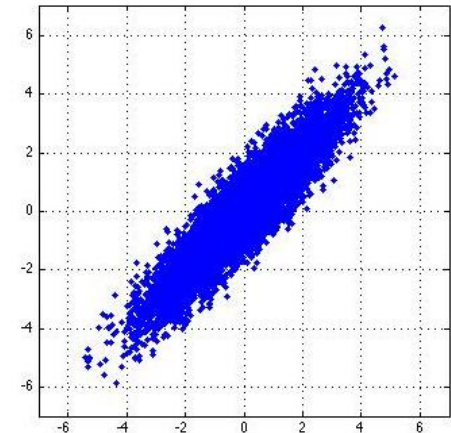
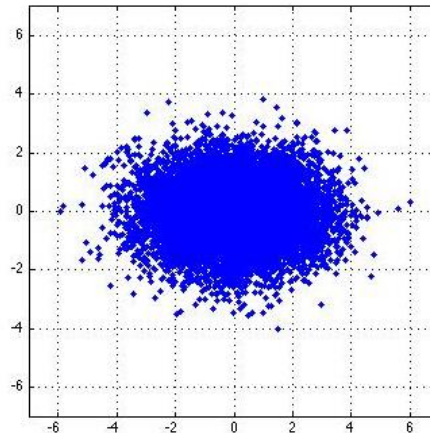
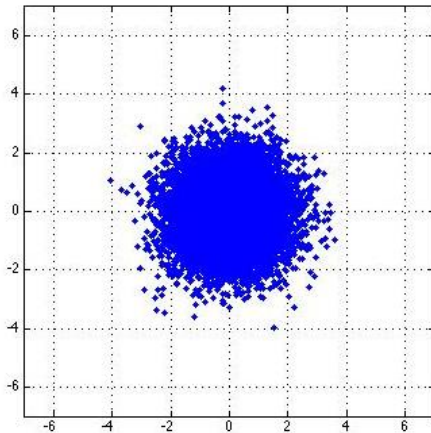


Joint (pixel) probabilities

- An image is parameterized as a vector, $\mathbf{j}$, of pixels
 - For each color channel
 - For now, we just look at luminance, greyscale
- How do we parameterize $P(\mathbf{j})$?
 - As a joint probability $P(j_1, j_2, \dots, j_N)$
 - Pixels are not independent $P(j_1, j_2) \neq P(j_1)P(j_2)$
 - How do we describe dependence?
- Gaussian variables
 - Zero covariance $\rightarrow$ independence

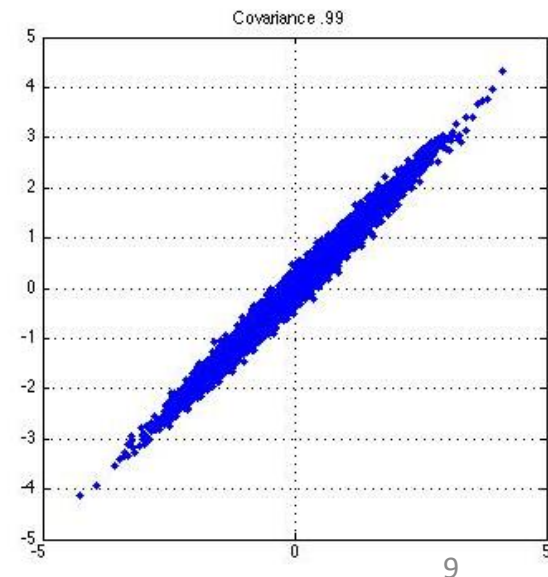
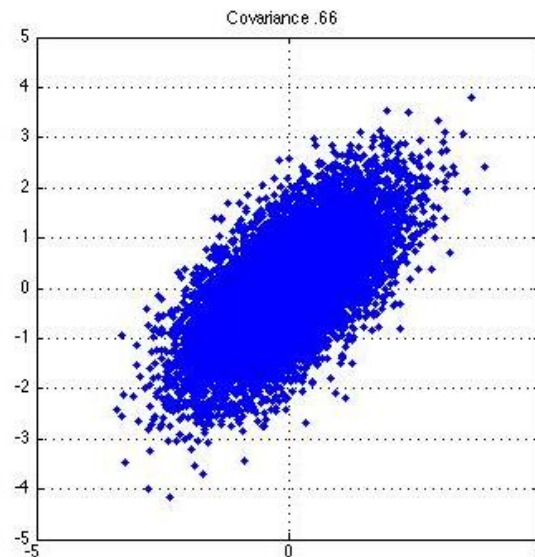
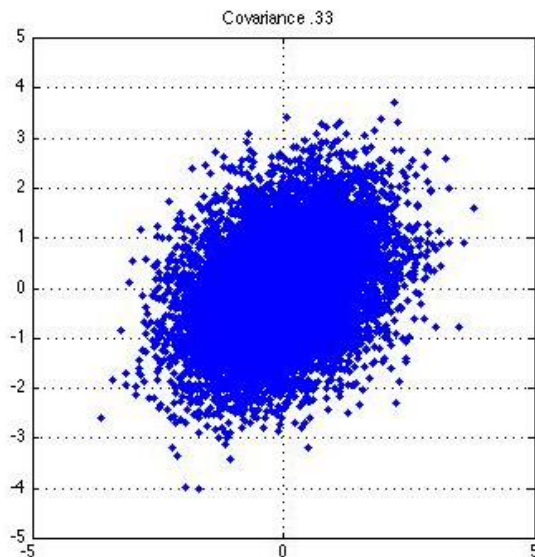
Covariance

- Which of the below data clouds have non-zero covariance?



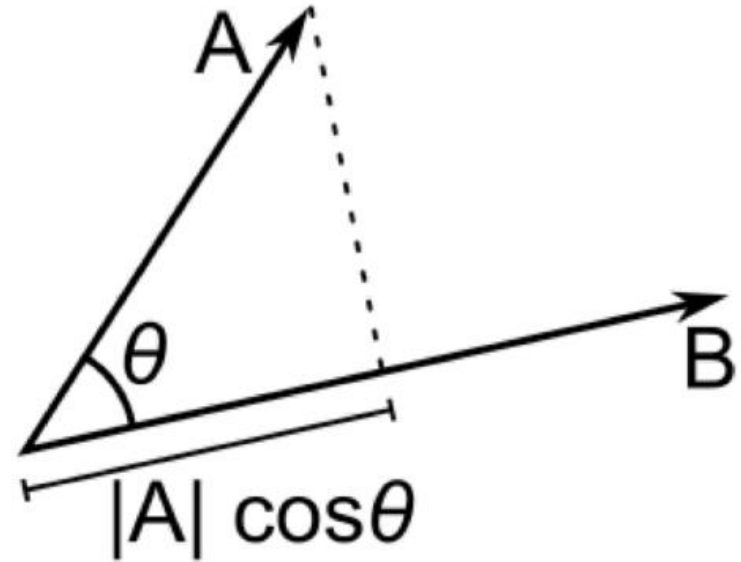
Covariance

- When covariance $\rightarrow 1$
 - The two variables start becoming identical
 - The 2D representation is redundant
 - A 1D representation would be sufficient



Vector projection

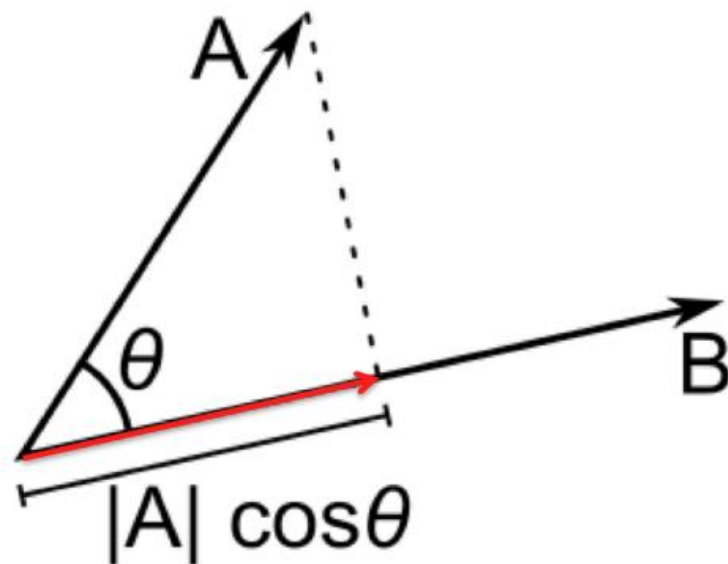
- 2D vector in a 1D basis
 - Projection or least squares approximation



- If the basis vector B has length 1 then
$$A \cdot B = |A||B|\cos(\theta) \Rightarrow A \cdot B = A^T B = |A|\cos(\theta)$$
- And the approximation of A in B's direction is $A \approx (A^T B)B$

Vector projection

- 2D vector in a 1D basis
 - Projection or least squares approximation



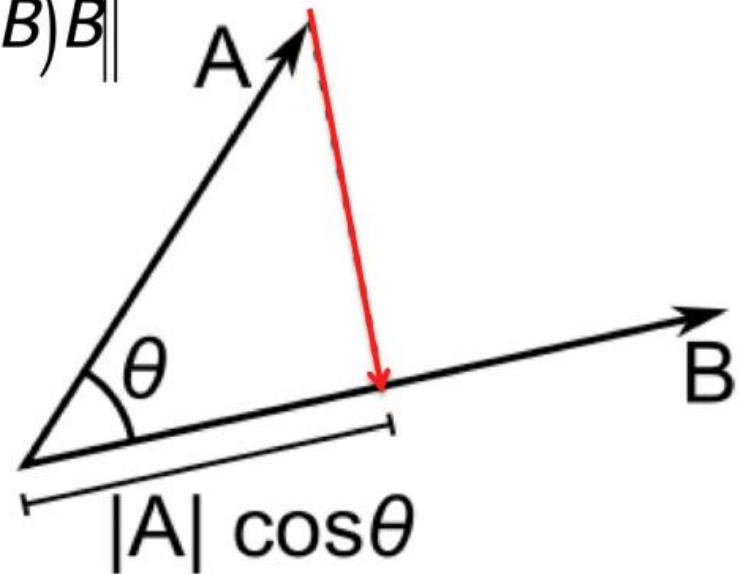
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Vector projection

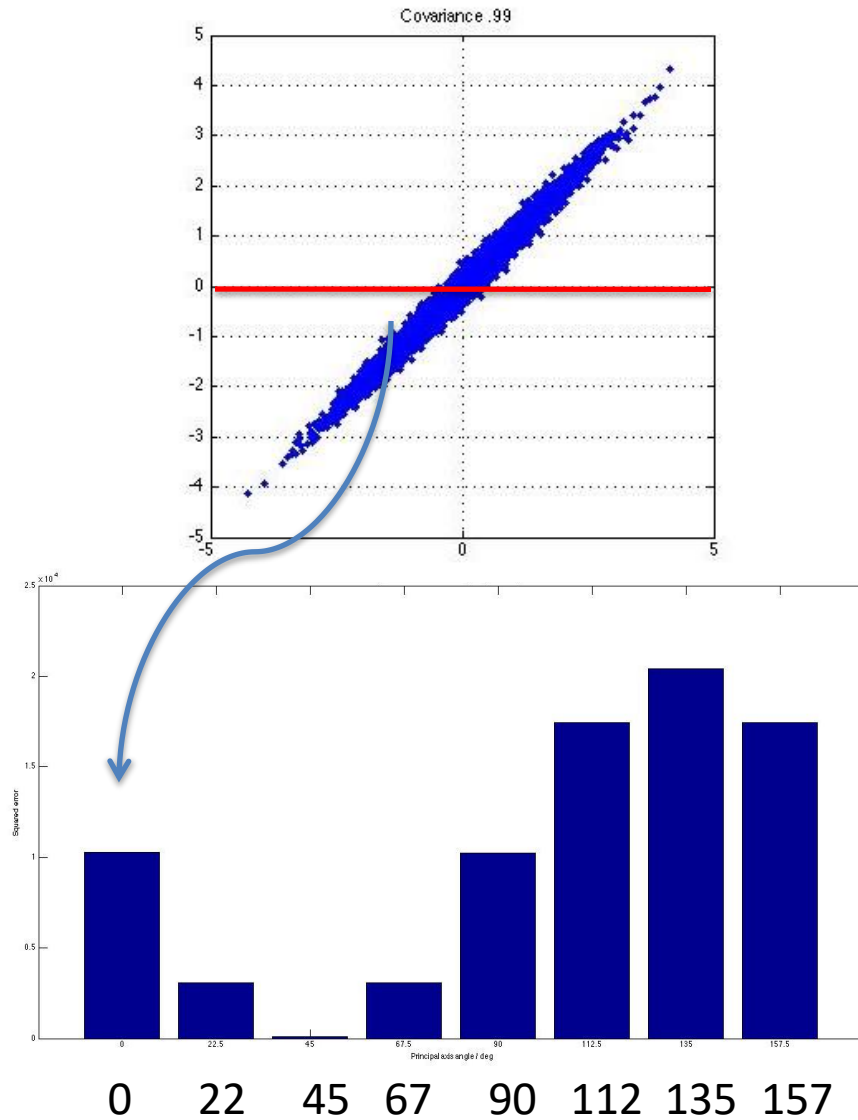
- How good is the approximation?

- Error is $A - (A^T B)B$

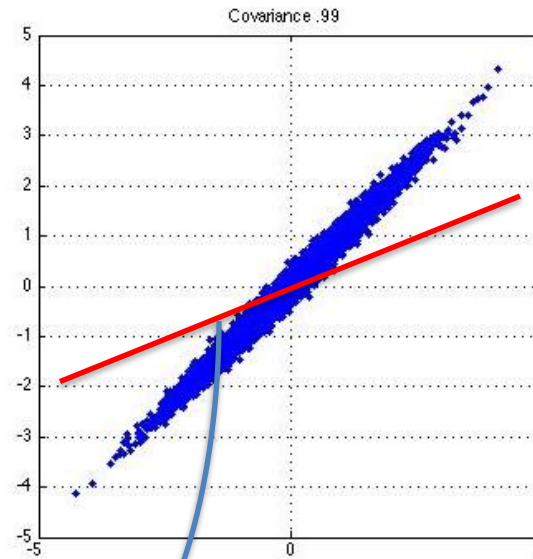
- Squared error is $\|A - (A^T B)B\|^2$



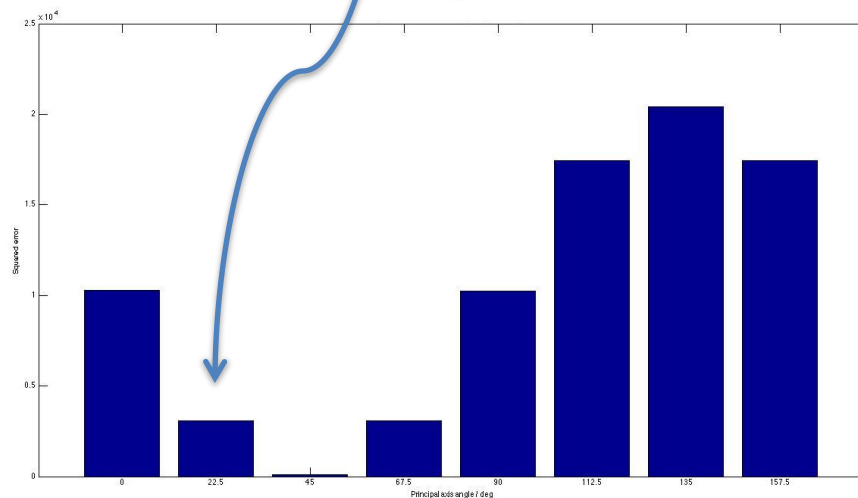
Minimizing the error



Minimizing the error



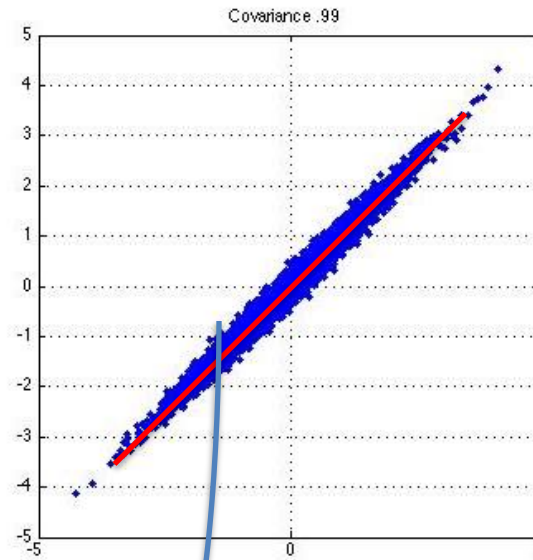
Squared Error



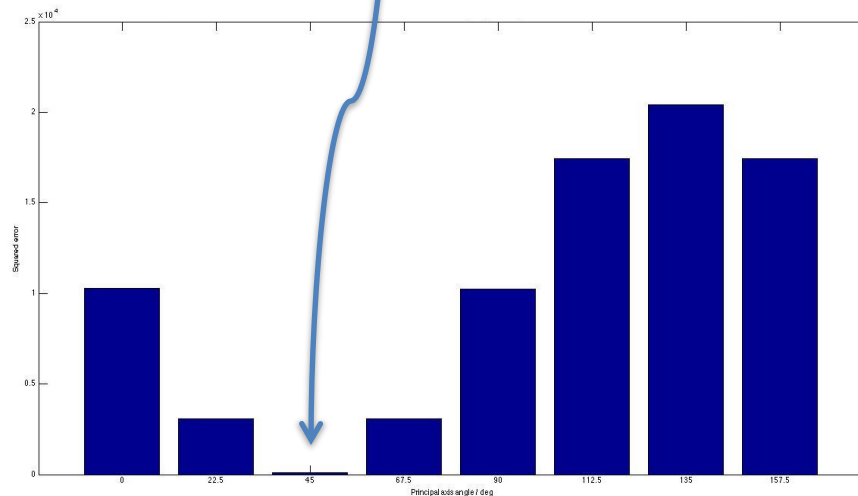
Degrees

0 22 45 67 90 112 135 157

Minimizing the error



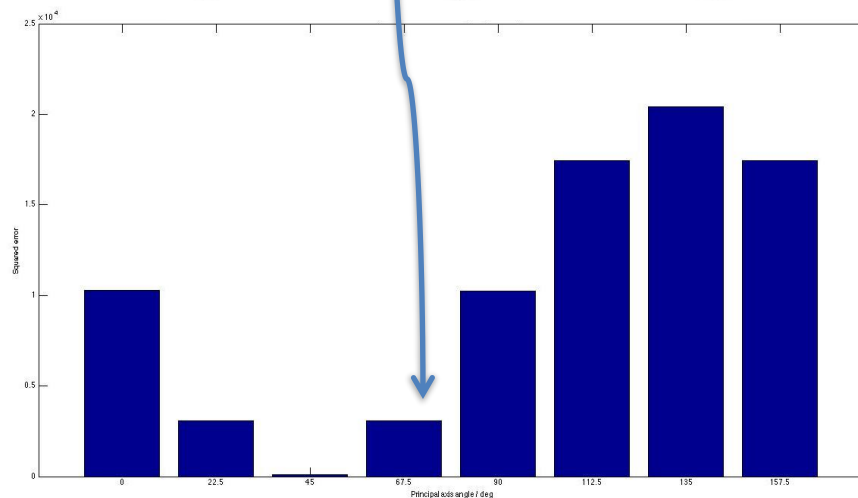
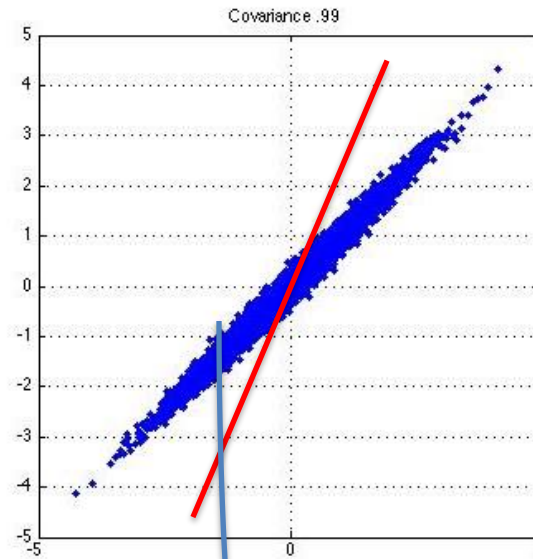
Squared Error



Degrees

0 22 45 67 90 112 135 157

Minimizing the error

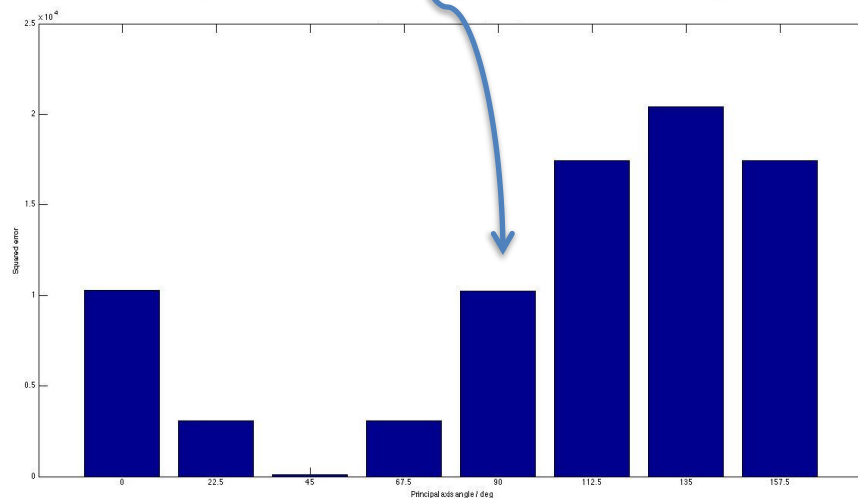
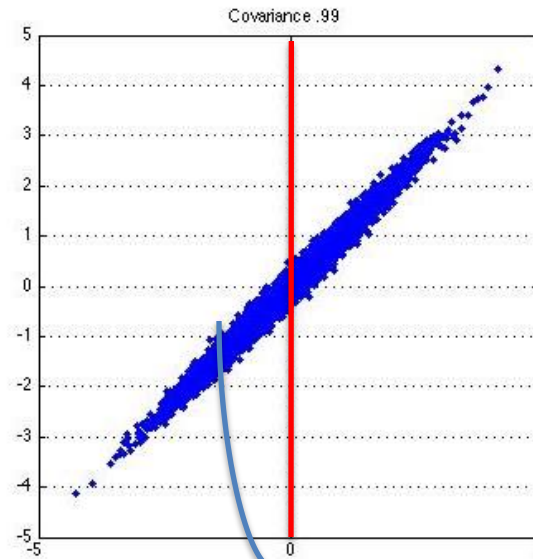


Squared Error

Degrees

0 22 45 67 90 112 135 157

Minimizing the error

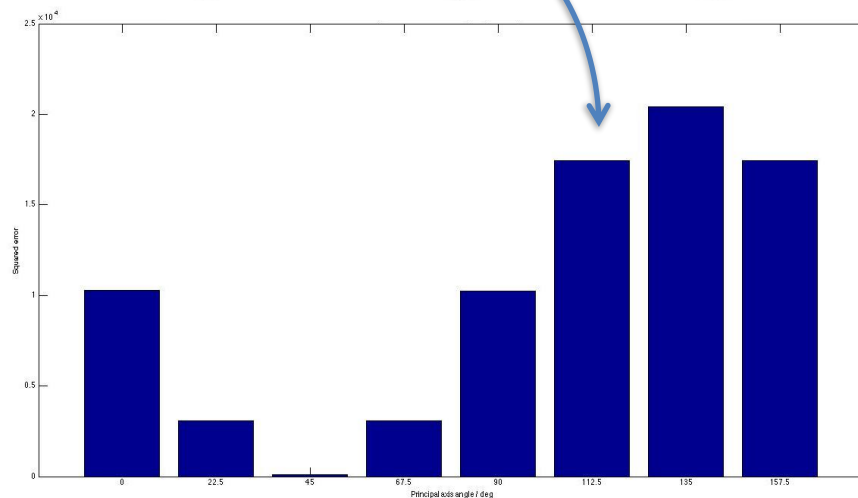
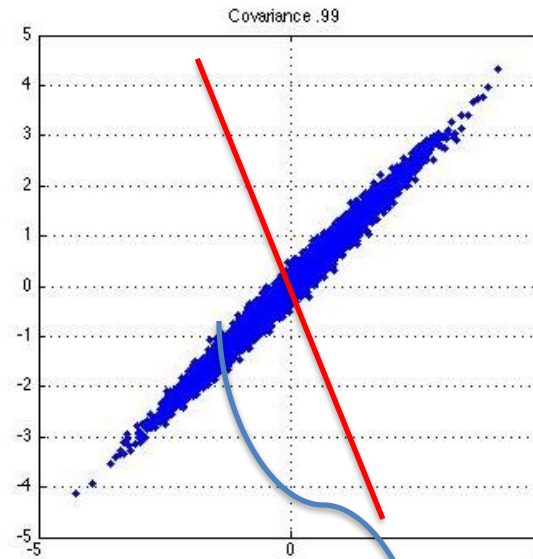


Squared Error

Degrees

0 22 45 67 90 112 135 157

Minimizing the error

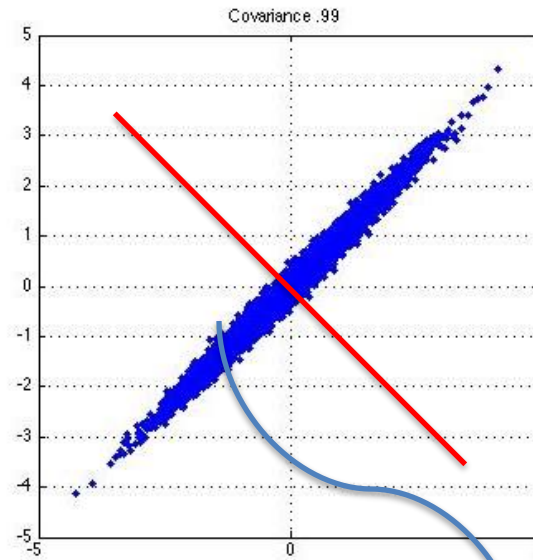


Squared Error

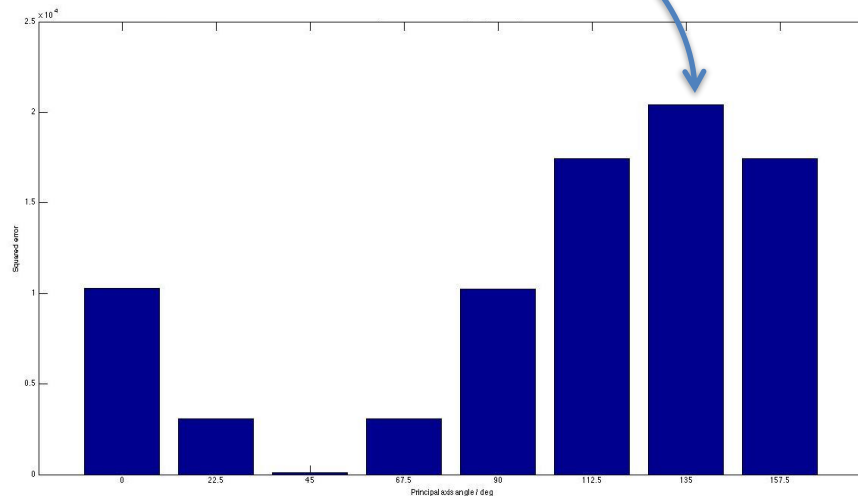
Degrees

0 22 45 67 90 112 135 157

Minimizing the error



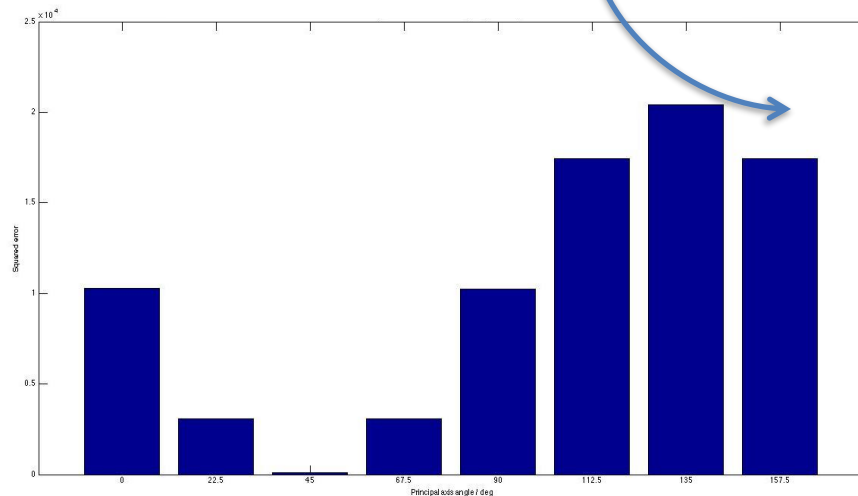
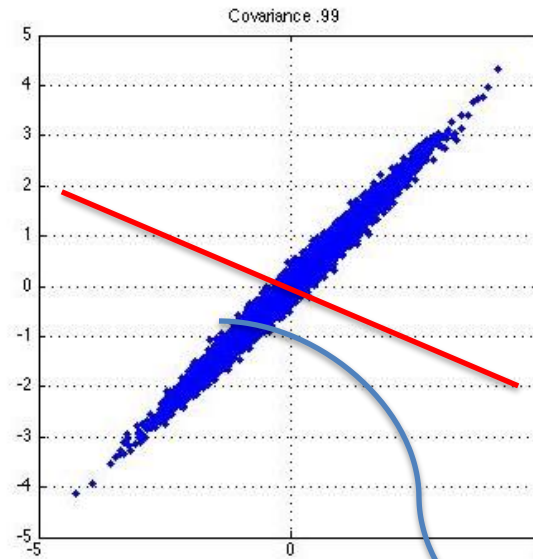
Squared Error



Degrees

0 22 45 67 90 112 135 157

Minimizing the error

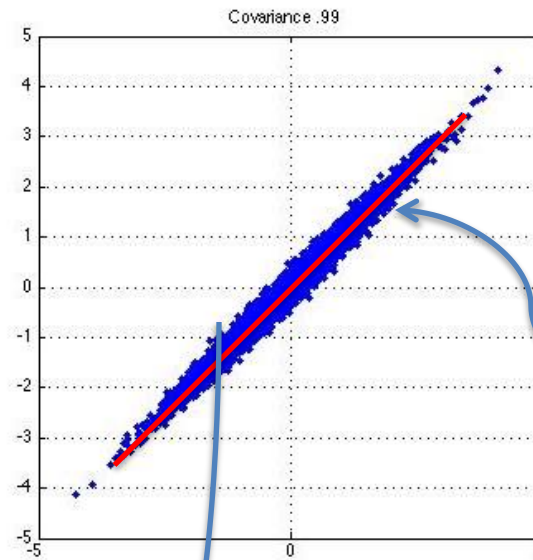


Squared Error

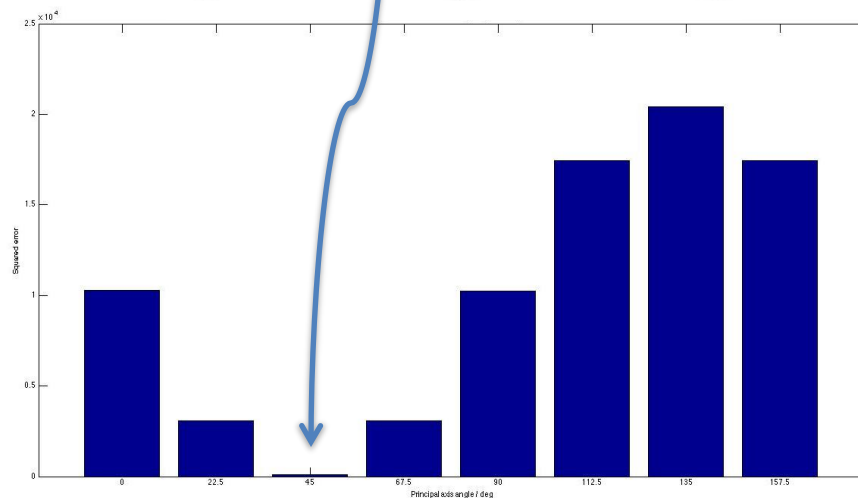
Degrees

0 22 45 67 90 112 135 157

Minimizing the error



First principal component

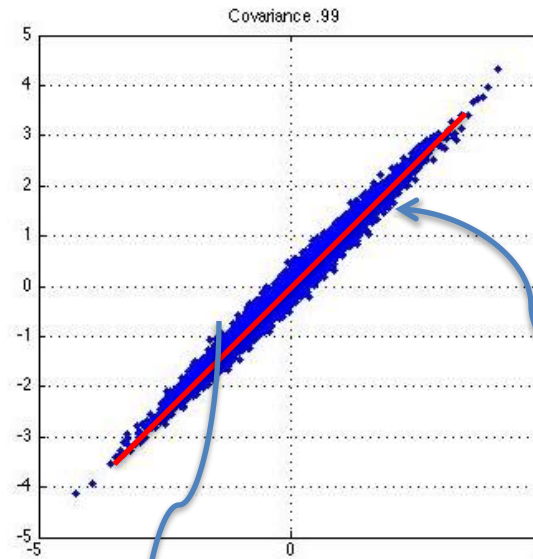


Squared Error

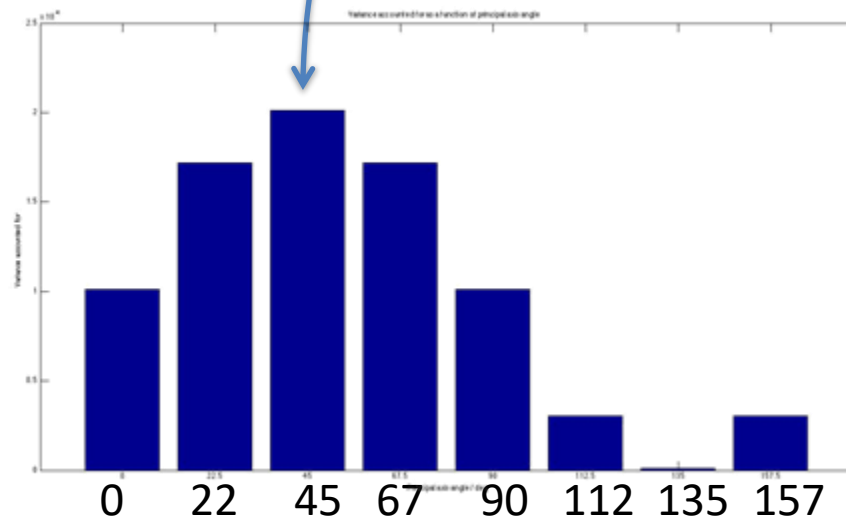
Degrees

0 22 45 67 90 112 135 157

Maximizing the variance *along* the principal component



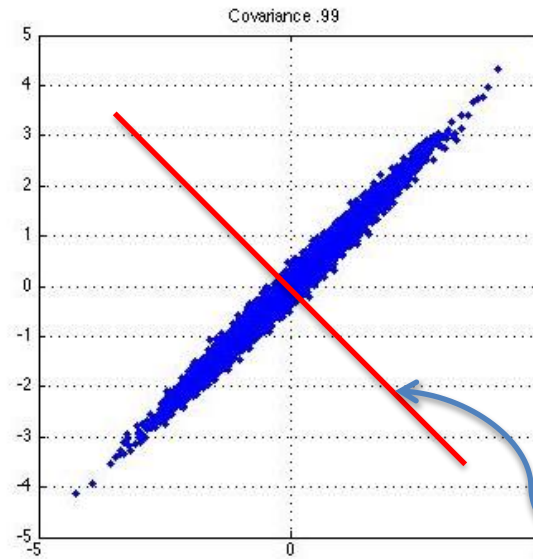
First principal component



Variance along PC1

Degrees

A principal component for the “error”



Second principal component

- Finding principal components is iterative
 - First find one, then the next
 - Trivial for 2D but not for many dimensions

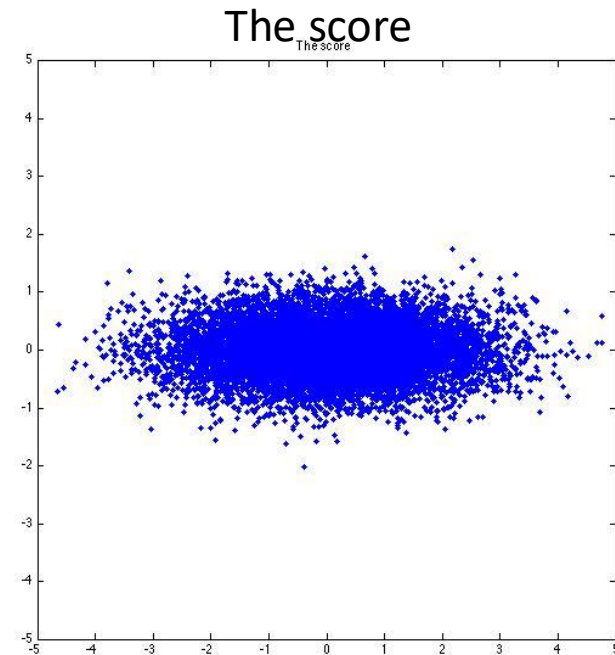
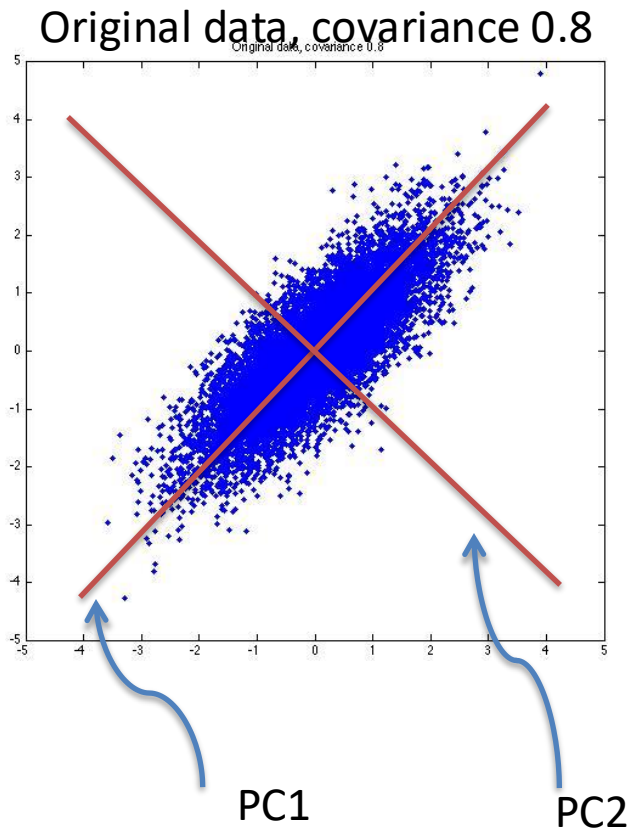
Principal component analysis (PCA)

- Find the principal components
 - N N -dimensional vectors (N -by- N matrix)
 - N is the number of measurement dimensions
 - The PC's are called the coefficients in Matlab
 - PC's are the eigenvectors of the covariance matrix
 - PC's are orthogonal – forms a new rotated basis
- Project the original data-set onto the PCs
 - This gives you the “score”

Principal component analysis (PCA)

- The PC's are ordered according to their variance
 - Gives an indication of their “importance”
 - The eigenvalues of the covariance matrix

Principal component analysis (PCA)



Notice greatest variance
along 1st dimension

Image PCA

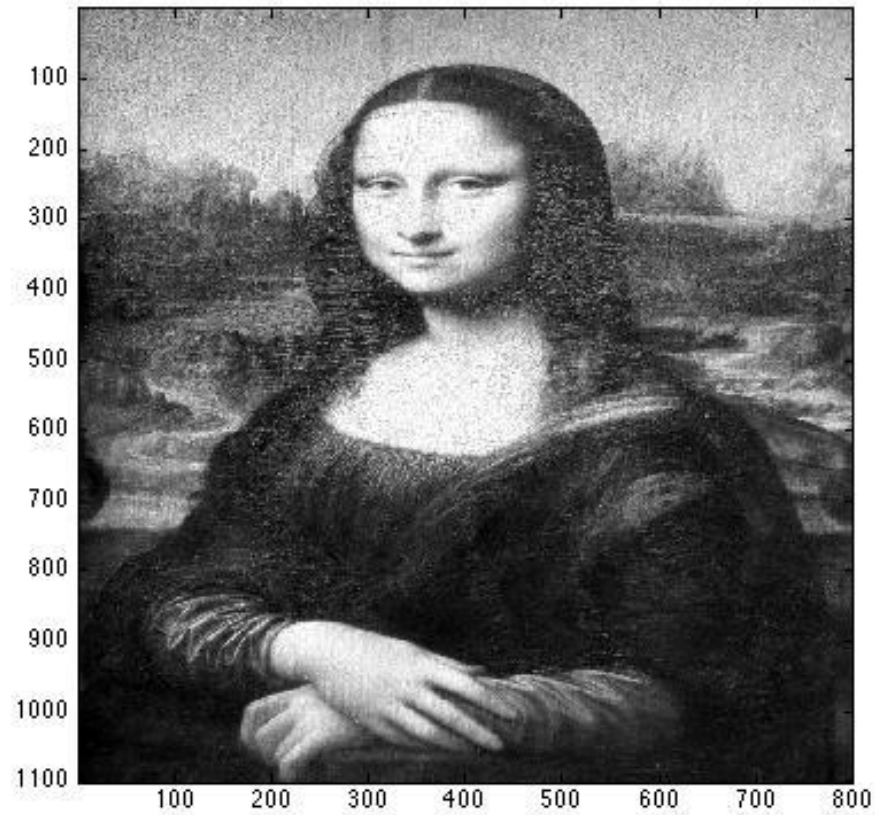
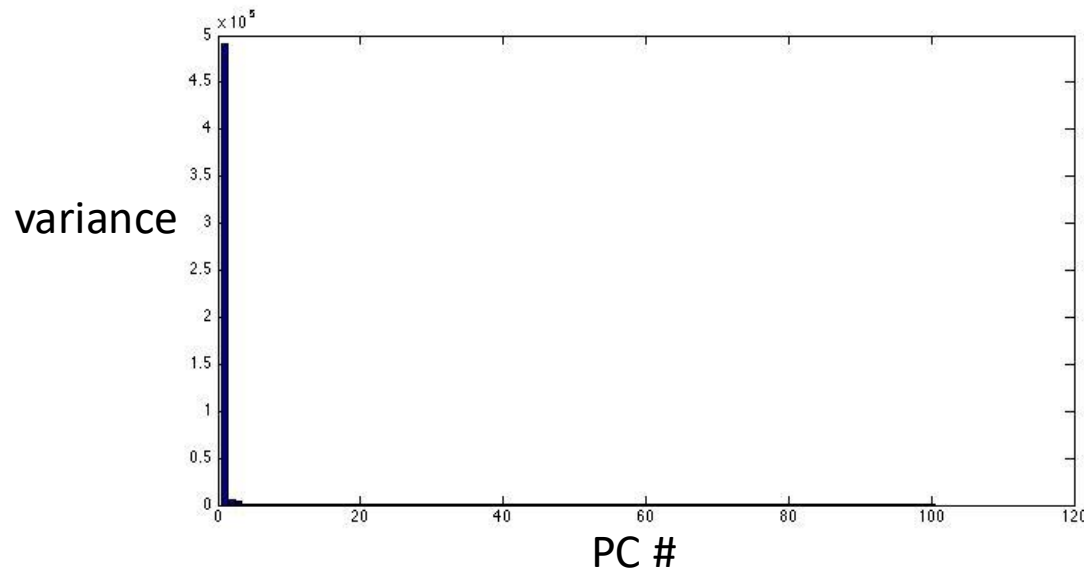


Image PCA

- Mona Lisa is just a bunch of samples from image patch space
- Given an 1100x800 pixel image
 - $110 \times 80 = 8800$ 10-by-10 patches
 - A 10-by-10 patch = 100-dimensional vector
- Enough for PCA
- How dimensions are relevant / informative?
 - How many neurons do we need?

Image PCA

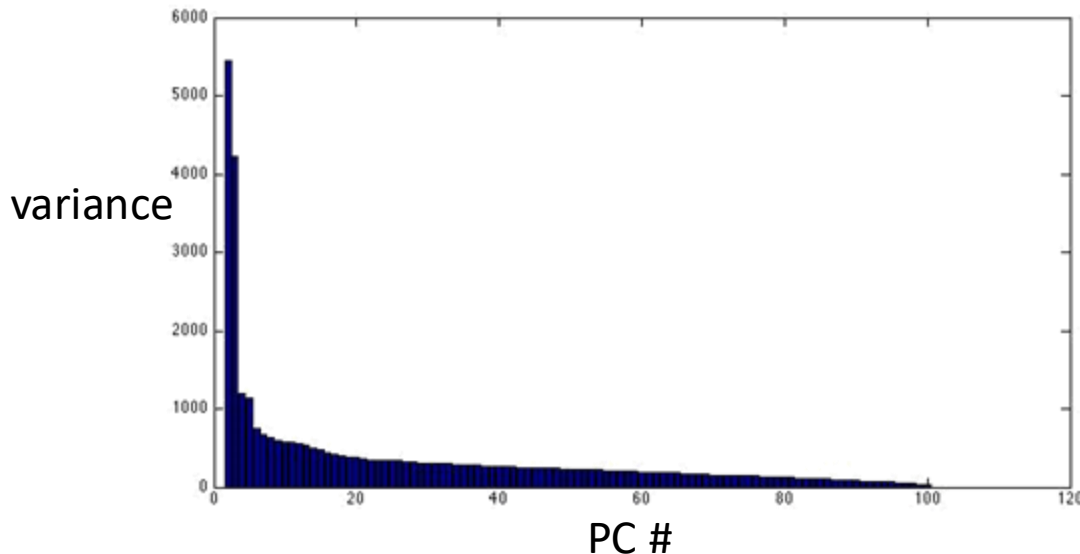
- Sorted eigenvalues



- Shows that PC1 dominates!
 - Let's take it out to have a closer look at the others...

Image PCA

- Sorted eigenvalues 2-100



- PC2 and PC3 also carries a lot of variance
- PC4-6 carries some variance
- But then it's just downhill

Image PCA

- What are these PCs? Let's have a look!

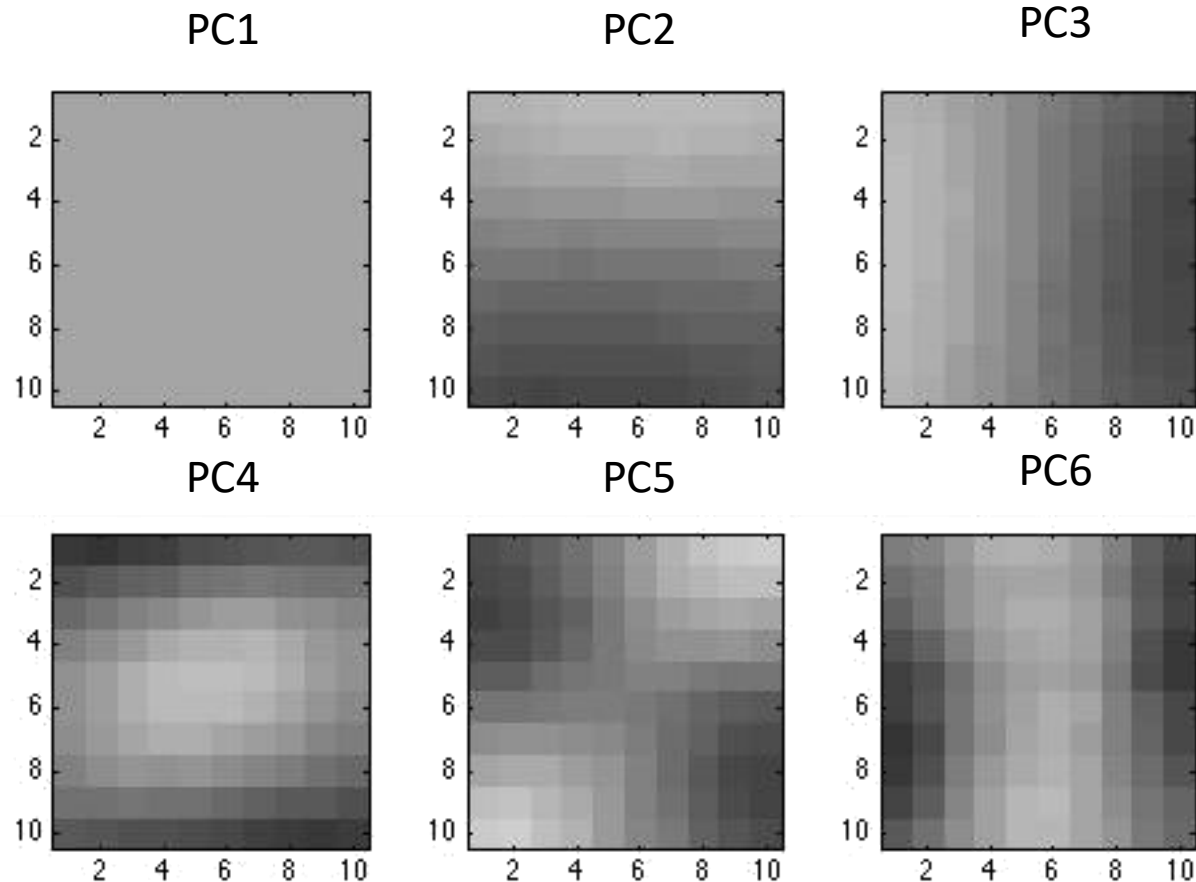


Image PCA

FIGURE 3.7

PCs? Let's have a look!

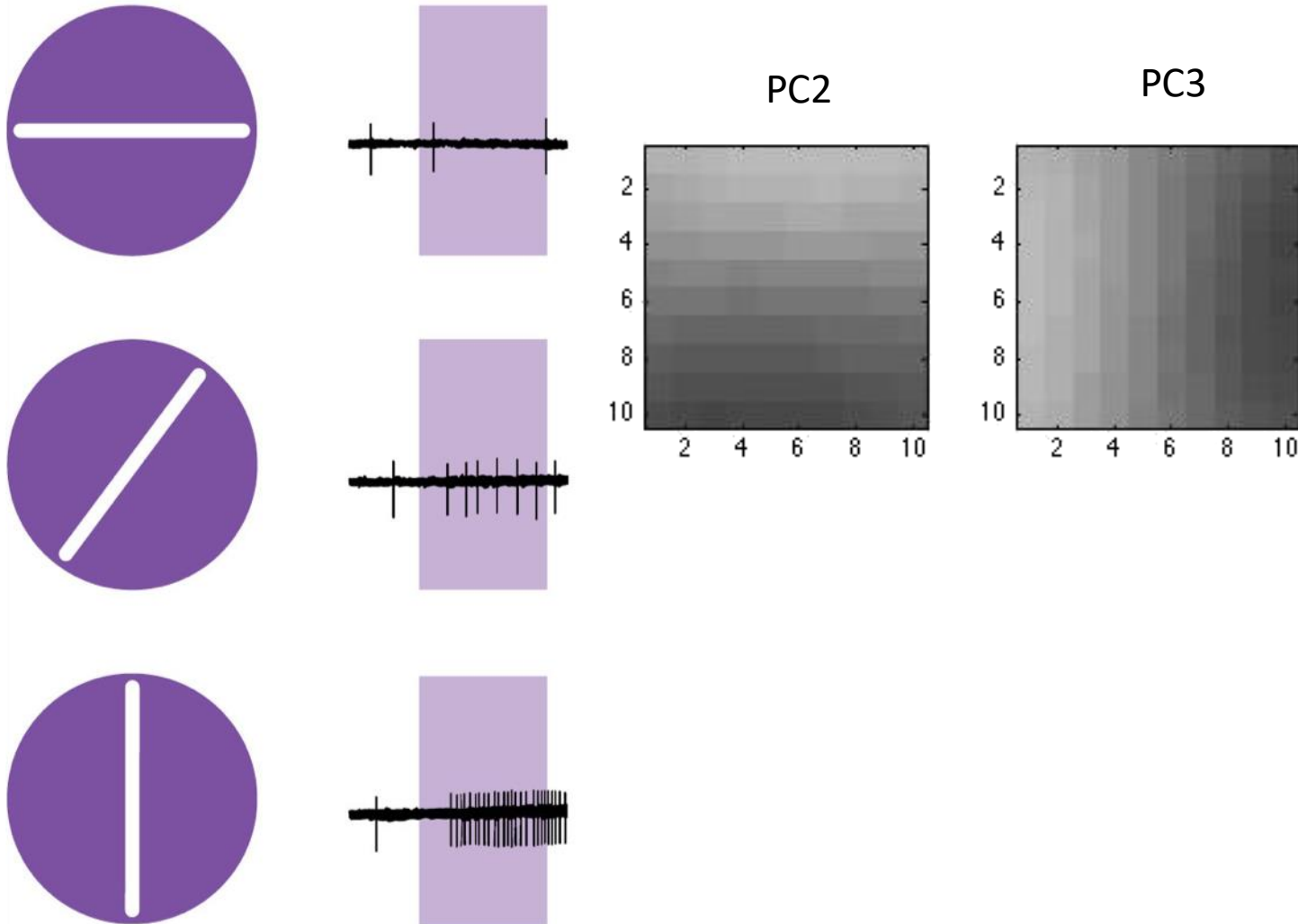


Image PCA

- What are these PCs? Let's have a look!

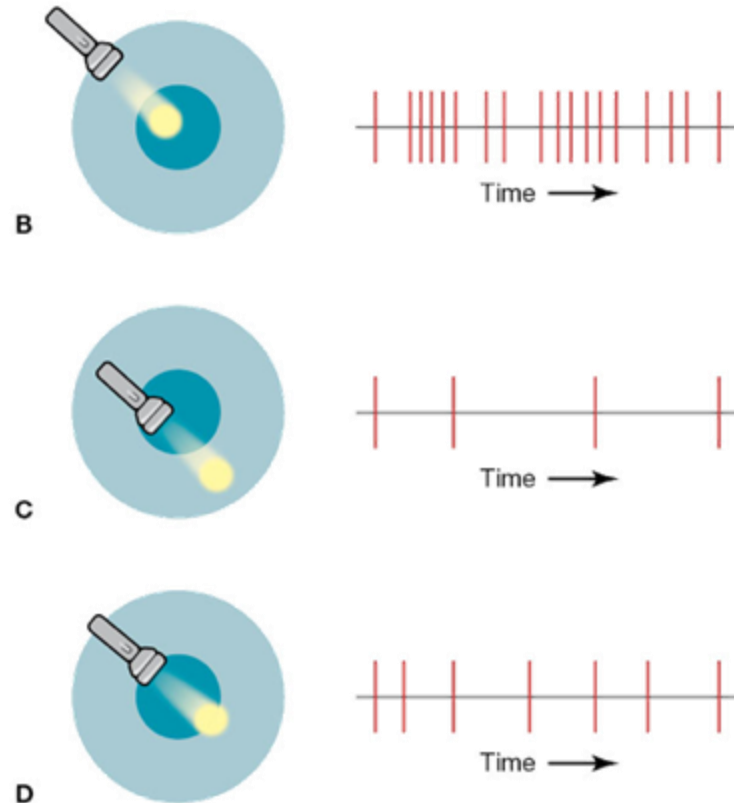
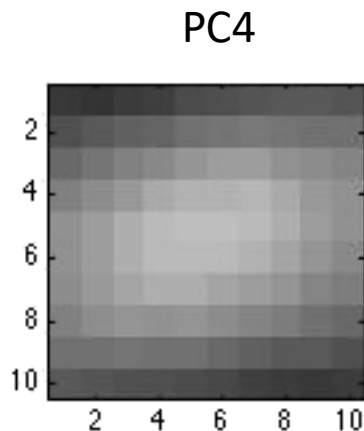


Image PCA

- What about the rest (94) PC's
- Do we need them?
- Let's reconstruct the image from the first 6 PCs

Image PCA

Who's who?

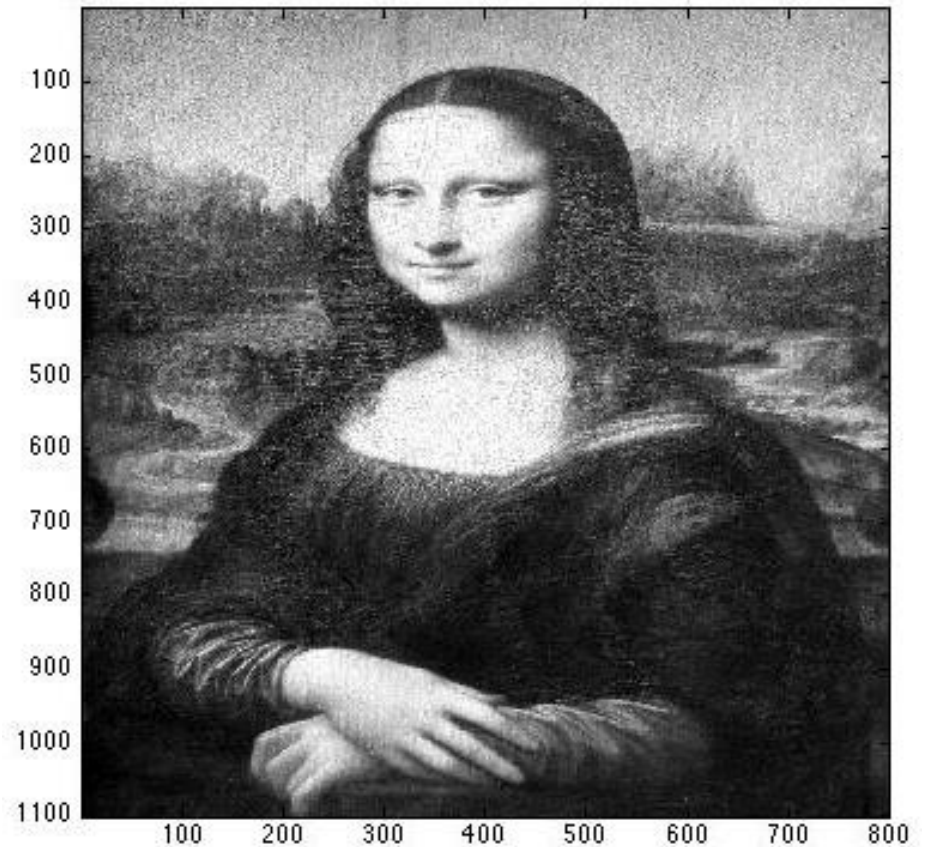
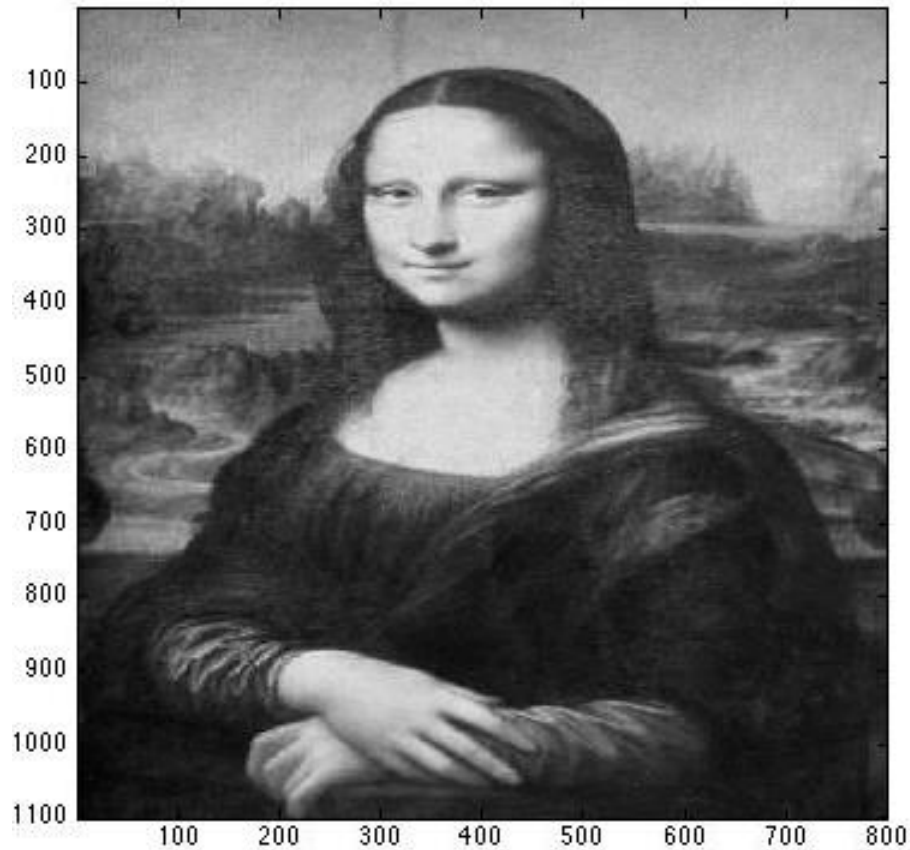


Image PCA

$$\mathbf{J} = \mathbf{A}\mathbf{W}^T$$

- $\mathbf{J}$ is the 8800-by-100 data matrix
- $\mathbf{A}$ is the 8800-by-100 data matrix in PC space
- $\mathbf{W}$ is a 100-by-100 component matrix
 - Each column is a 100-dimensional component
- Reconstruct the n^{th} image
 - Take the n^{th} row of $\mathbf{A}$

Image PCA

$$\mathbf{J} = \mathbf{A}\mathbf{W}^T$$

- $\mathbf{J}$ is the 8800-by-100 data matrix
- $\mathbf{A}$ is the 8800-by-100 data matrix in PC space
- $\mathbf{W}$ is a 100-by-100 component matrix
 - Each column is a 100-dimensional component
- Reconstruct the n^{th} image
 - Take the n^{th} row of $\mathbf{A}$
- Using the m^{th} principal component
 - Take the m^{th} column of $\mathbf{A}$, the m^{th} column of $\mathbf{W}$

Image PCA



Figure 3.5: Left: A detail from the original image Mona Lisa in Figure 3.2. Center: The same detail made of image patches reconstructed using six principal components. Right: The same detail reconstructed using only one principal component. See main text for details..

Image PCA

- PCA
 - Provides a prioritized list of “sensors”
 - Denoises
 - Compress images
 - Provides a superior representation of images
 - Compared to a pixel-by-pixel representation
 - A candidate for image components

Technical note on standardisation

- Before PCA
 - Subtract the mean *across patches*
 - Remember to add it back after PCA
 - Do not divide by standard deviation
 - All dimensions (pixels) are comparable

Now to the exercises..